

SELECTED PUBLICATIONS (CONTINUED)

Toward Less Conservative Distributed Stability Analysis of Power Systems via Matrix-Valued Differential Passivity Indices

Xi Ru, Cong Fu, **Zhongze Li**, Xiaoyu Peng, Feng Liu.
arXiv:2605.04821, 2026.

Compositional Grid Codes With Guarantee on Both Stability and Dynamic Performance

Xiaoyu Peng, Cong Fu, **Zhongze Li**, Xi Ru, Zhaojian Wang, Feng Liu.
IEEE Transactions on Power Systems, 2026.

Positive Damping Region: A Graphic Tool for Passivization Analysis with Passivity Index

Xiaoyu Peng, Xi Ru, **Zhongze Li**, Jianxin Zhang, Xinghua Chen, Feng Liu.
arXiv:2601.10475, 2026.

Relative-Degree Wall Restricts Passivity-Based Stability Analysis in Inverter-Dominant Grids

Xiaoyu Peng, **Zhongze Li**, Xi Ru, Xinghua Chen, Feng Liu.
arXiv preprint, 2026.

SELECTED ACADEMIC ACTIVITIES

Presenter, The 4th Energy Conversion and Economics Annual Forum (ECE 2024), Beijing, China. Talk: "Distributed Stability Analysis for Power Systems with Grid-Following and Grid-Forming Devices."

RESEARCH METHODS & TOOLS

Robust control and frequency-domain analysis: H-infinity methods, structured singular value (μ), passivity/dissipativity, Nyquist/Bode analysis, uncertainty modeling.

Power-system modeling and simulation: converter-dominated systems, EMT/RMS multi-timescale modeling, small-signal and nonlinear stability analysis.

Computational tools: MATLAB/Simulink, Python; numerical simulation and reproducible research workflows.

LINKS

Academic homepage: <https://lizzz766.github.io>

Google Scholar: <https://scholar.google.com/citations?user=HKlw3YUAAAAJ&hl=en>

GitHub: <https://github.com/lizzz766>

Last updated September 2026. For the most up-to-date publication list, please refer to Google Scholar.